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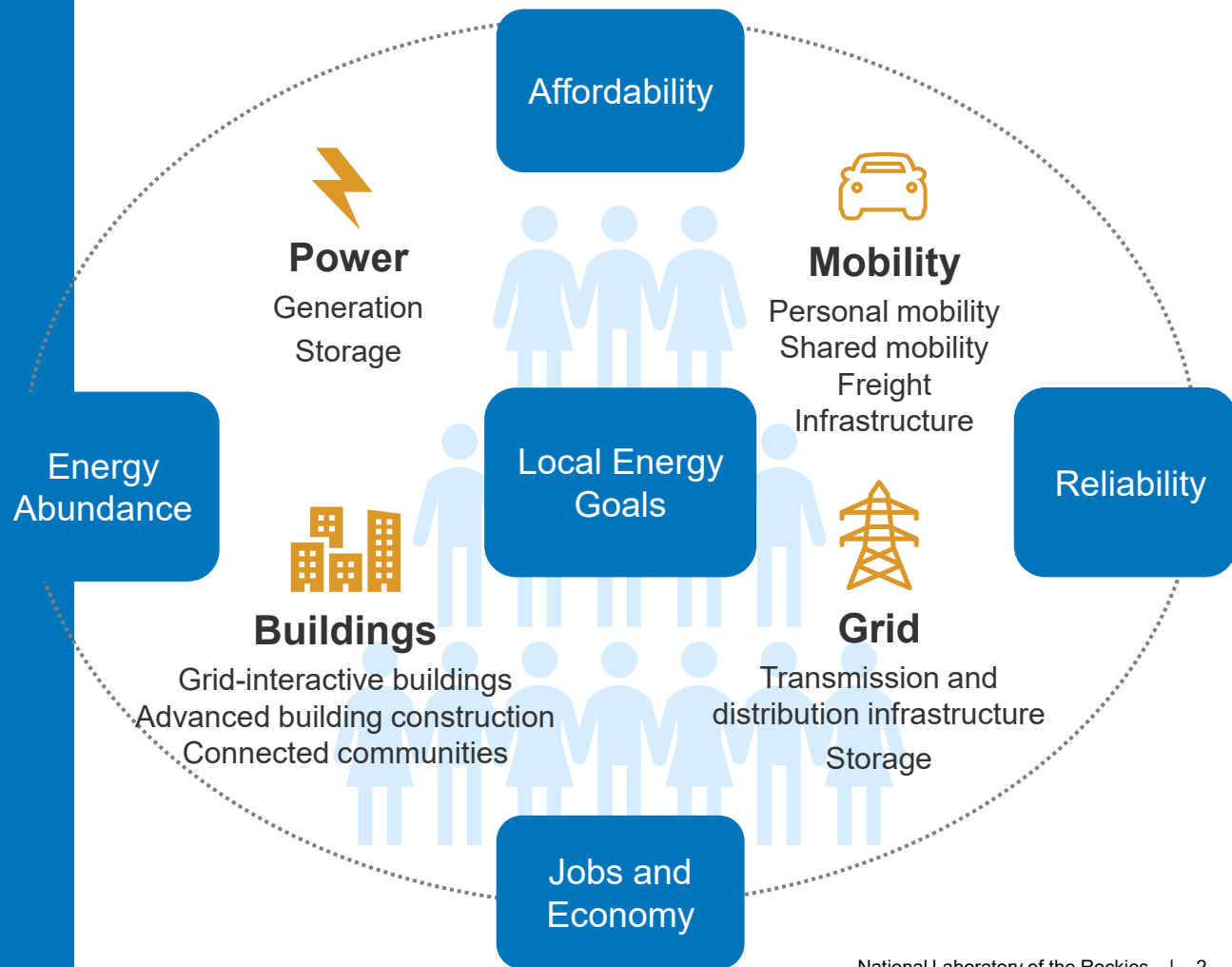
Preliminary Assessment of Distributed Energy Options in Greenville, Florida

Energy to Communities Expert Match

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May 2026

Energy to Communities (E2C) provides innovative, cross-cutting technical analysis to support local efforts to secure abundant, reliable, and affordable energy systems.



E2C Technical Assistance Opportunities



In-Depth Partnerships ~3 years

Multiyear partnership comprising teams (local governments, community-based organizations, and electric utilities) that work alongside national laboratory staff to apply robust modeling and analysis tools and simulate hardware in conditions that match communities' unique contexts to evaluate potential scenarios and strategies before full technology deployment.

Supports nine communities currently.



Peer-Learning Cohorts ~6 months

A total of 6 months of regular convenings where cohorts of up to 15 participants exchange strategies and best practices, learn in a collaborative environment, and workshop plans and strategies to overcome challenges relating to a similar energy topic. Eligibility varies by cohort topic.

Supports about 100 communities per year.



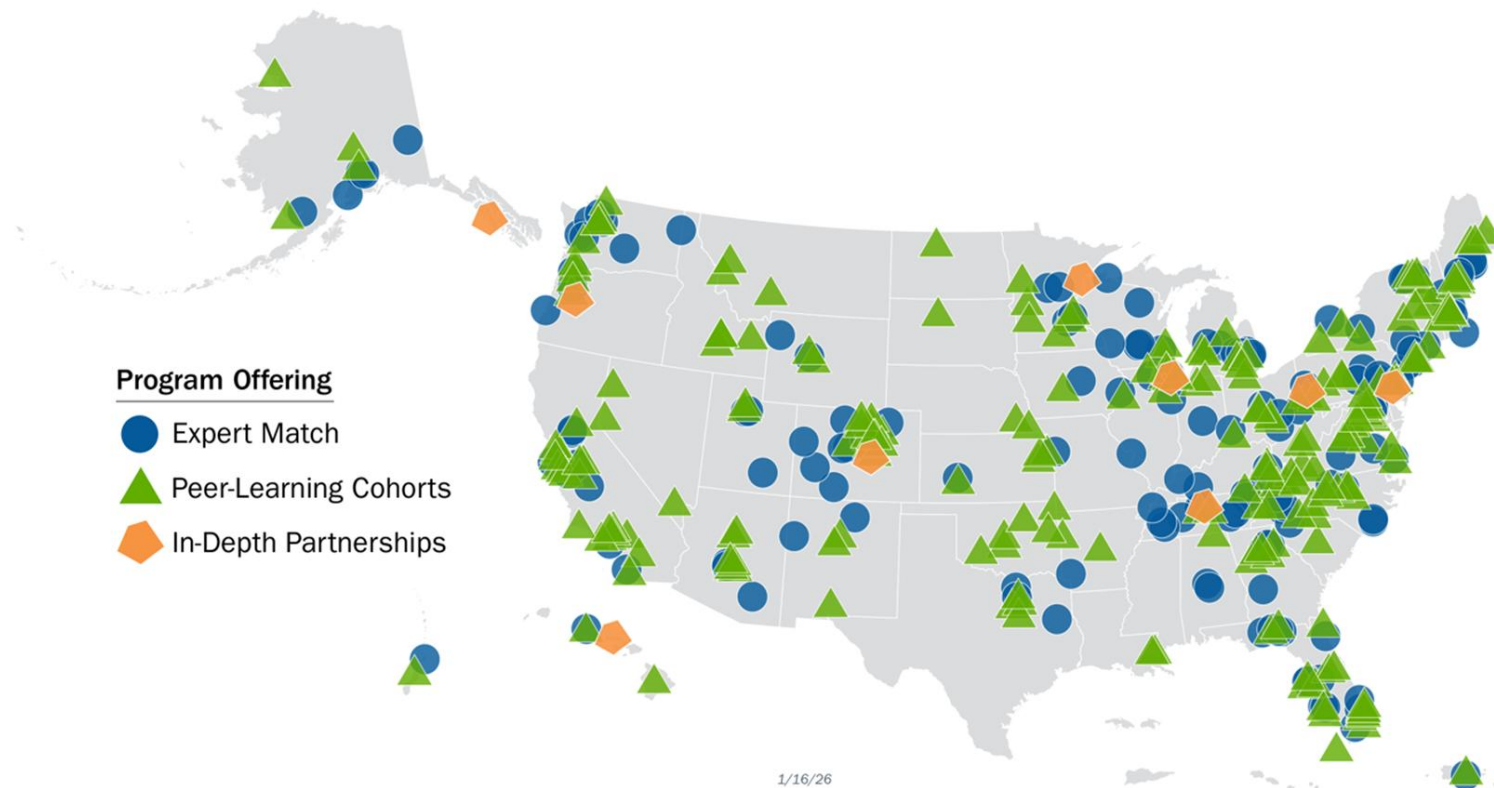
Expert Match ~3 months

Short-term, no-cost technical assistance for communities seeking to answer near-term energy questions. Eligible entities include local governments, utilities, community-based organizations, schools, and nonprofits.

Supports about 200 communities per year. Applications are accepted all year on a rolling basis.

www.nlr.gov/e2c

E2C Program Participants



1/16/26

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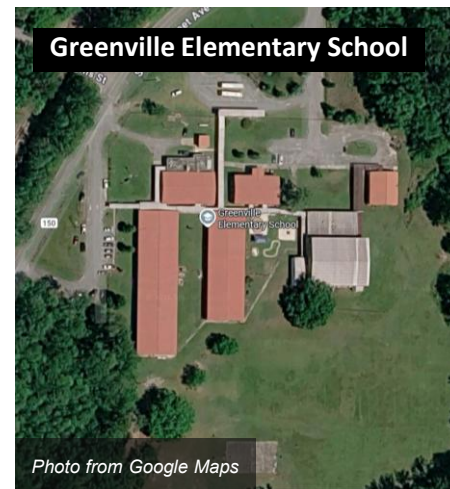
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Background and REopt Overview

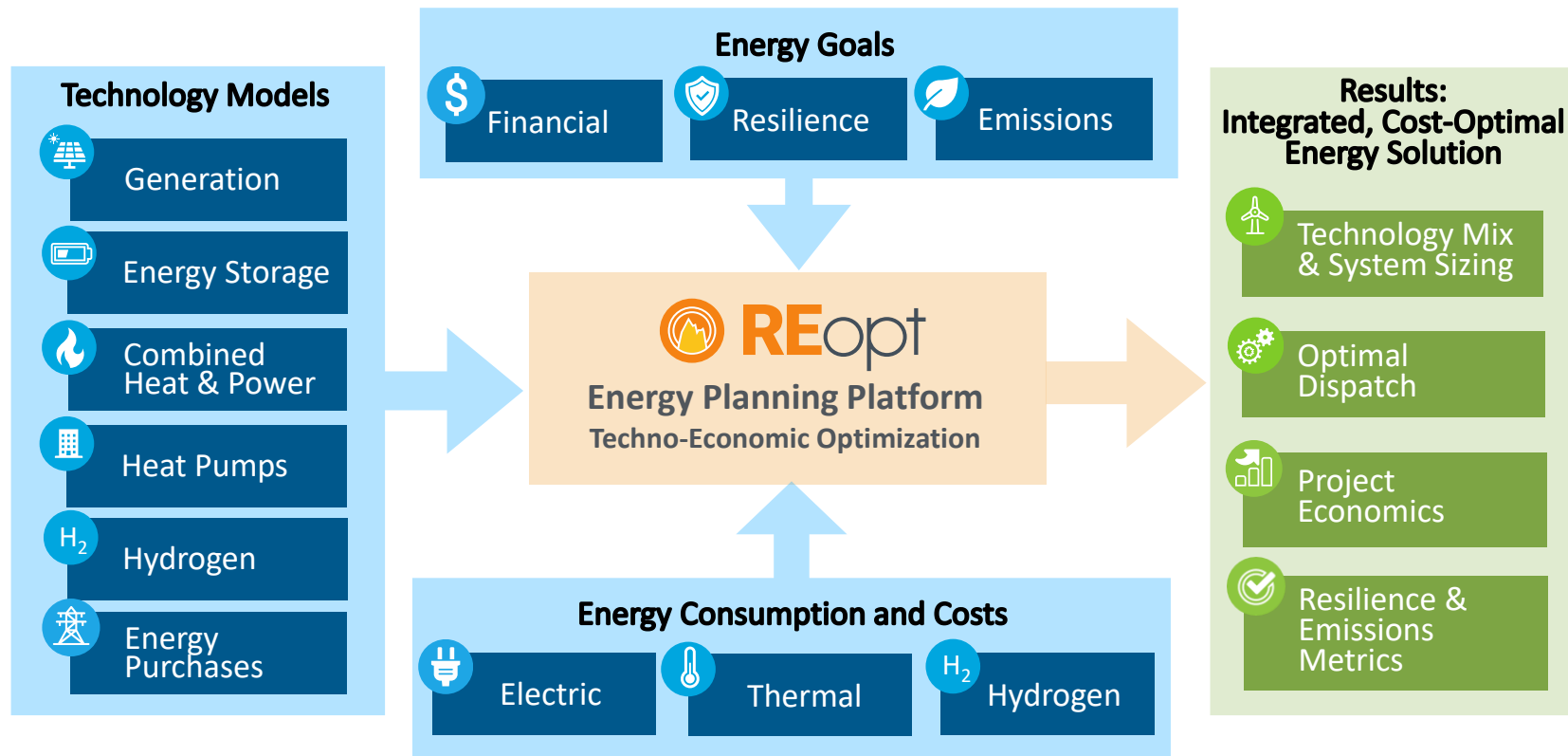
Analysis Overview

- NLR used the **REopt**[®] platform to evaluate the techno-economic potential of adding solar photovoltaics (PV), wind power generation, battery energy storage, and diesel generators at two sites in Greenville, Florida:
 - The Town of Greenville Grocery Store:**
1344 SW Grand Street, Greenville, FL
 - Greenville Elementary School:**
729 SW Overstreet Ave., Greenville, FL
- The initial analysis focuses on the ability of these technologies to reduce **electricity costs** and increase the sites' ability to withstand **outages**.



REopt Overview

REopt is accessible via user-friendly web tool and open-source code.



REopt Minimizes the Life Cycle Cost of Energy

- **Life cycle cost (LCC)** of energy: The present value of all costs of energy at the site throughout the analysis period.

$$\text{LCC} = \text{Capital Costs} - \text{Available Incentives} + \text{O\&M Costs} + \text{Electric Grid Purchases} + \text{Fuel Purchases}$$

- **Net present value (NPV)** of DER system: The life cycle cost savings (difference in LCC) between the business-as-usual (BAU) case and the optimized (OPT) case.

$$\text{NPV} = \text{LCC}_{\text{BAU}} - \text{LCC}_{\text{OPT}}$$

REopt identifies the **life cycle cost-optimal** DER system that achieves the site's energy goals.

If $\text{NPV} > 0$, the project provides cost savings relative to the BAU case.

If $\text{NPV} < 0$, the project is more expensive than the BAU case.

Caveats

Topic	Caveat
Model inputs	This analysis uses a variety of assumptions regarding the cost of technologies, future loads, and future energy tariffs. These assumptions may differ from reality. Additional financial analysis may be required to more accurately model costs and benefits.
Perfect foresight	The modeled battery dispatch strategy assumes the system can perfectly predict future electric loads and can thus be considered an upper bound on performance.
Preliminary analysis	These initial findings should be considered preliminary and can serve as the basis for further viability analyses of solar and storage.
Electricity system upgrade costs for resilience	The analysis does not include cost estimates for upgrading the electrical system for islanded operation during an outage (e.g., critical load panel, microgrid controller). Actual upgrade needs and costs are site specific and require additional analysis.
Building load	Future load changes (e.g., energy efficiency upgrades or electric vehicle charging stations) were not considered in this analysis. Future load growth may impact requirements for system sizing and resilience design.
Incentives	Incentives and eligibility for incentives may change in the future.
Interconnection upgrade costs	The project may incur costs for potential upgrades to the utility distribution system, which the utility may pass on to the project developers. These costs were not included in this analysis because of uncertainty and potential variability in these costs, which could be estimated later in the project development process.
Renewable generation resource data	The estimated power output from the solar PV and wind system is based on historical resource data and represented as typical meteorological year data. Actual power output may differ.
Project financing	The techno-economic model assumes all capital costs are paid upfront. Costs associated with project financing (such as interest on loans) is not included in this analysis.

Inputs and Assumptions

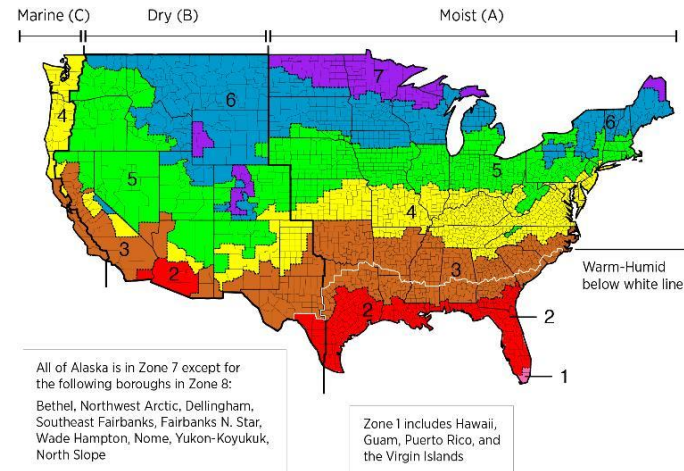
Modeling Hourly Loads

- Hourly interval data were estimated using the U.S. Department of Energy commercial reference buildings dataset¹:
 - This dataset comprises **16 building types** representing about **70%** of commercial buildings in the United States.
 - The modeled load uses the **Supermarket** profile to represent the Greenville Grocery store and the **Medium Office** profile to represent Greenville Elementary School, which will operate as a community center five days per week.
- Building loads are adjusted for the characteristics of the appropriate climate zone² (2A).

¹ <https://www.energy.gov/eere/buildings/commercial-reference-buildings>

² <https://basc.pnnl.gov/images/iecc-climate-zone-map>

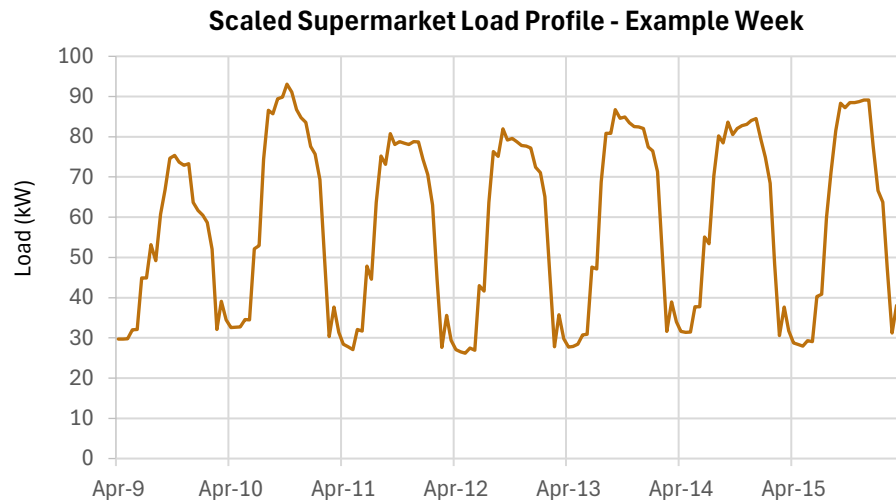
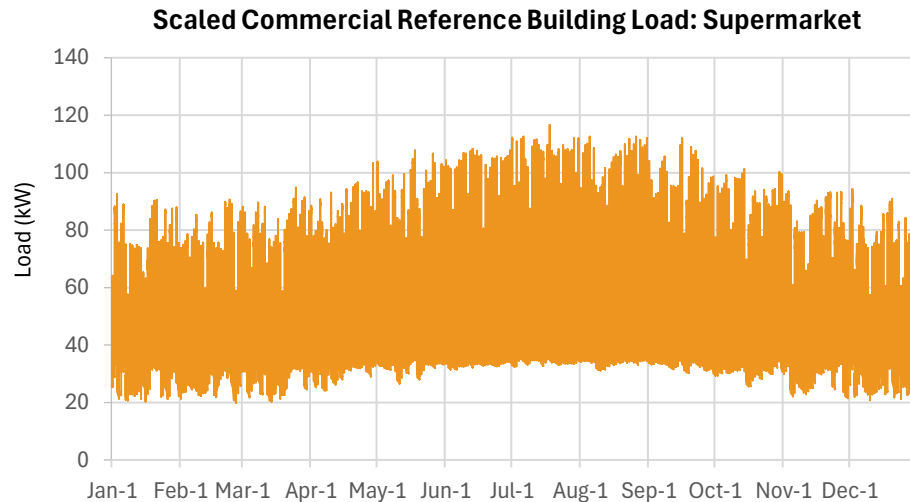
Building Types	
Large Office	Supermarket
Medium Office	Quick Service Restaurant
Small Office	Full-Service Restaurant
Warehouse	Hospital
Stand-alone Retail	Outpatient Health Care
Strip Mall	Small Hotel
Primary School	Large Hotel
Secondary School	Midrise Apartment



Modeled Grocery Store Load

- The load profile is obtained by scaling the Commercial Reference Building's annual energy consumption by the ratio of the actual building's square footage (**10,860 square feet [sq ft]**) and the reference building's square footage (**45,000 sq ft**).
- The modeled load peaks in the **summer** months and does not show a significant difference in consumption patterns between weekdays and weekends throughout the year.
- Daily demand is highest between approximately **8:00 a.m. and 8:00 p.m.**

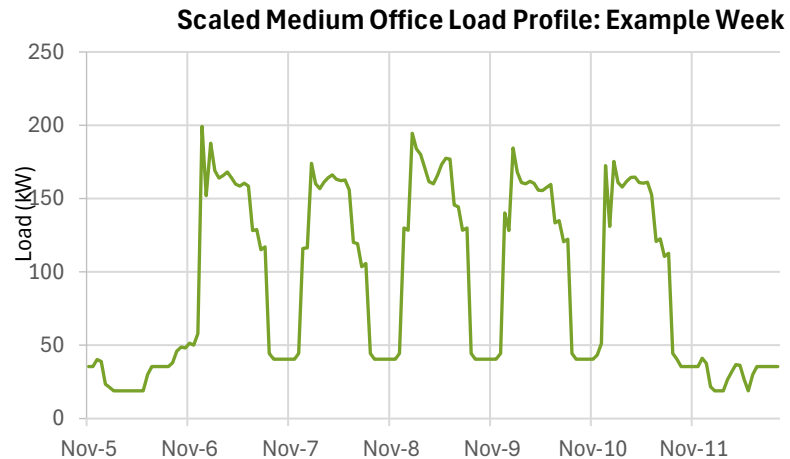
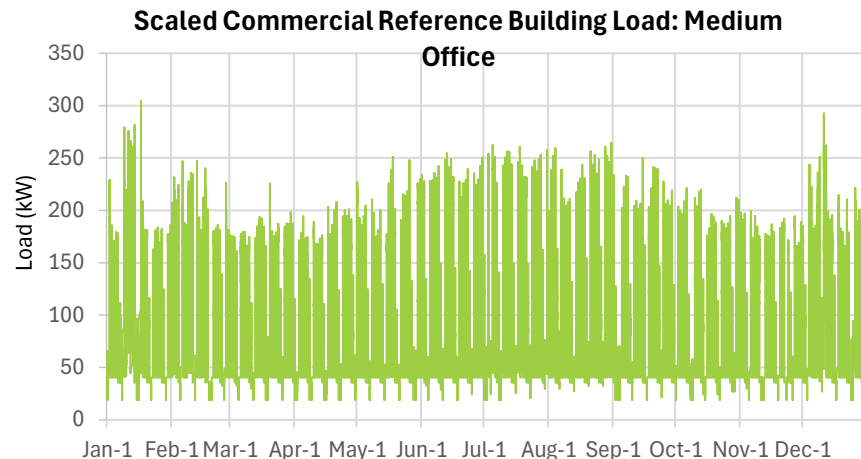
	Supermarket Profile
Peak load	116.6 kilowatts (kW)
Average load	61.3 kW
Minimum load	19.8 kW
Annual consumption	537,031 kilowatt-hours



Modeled Elementary School Load

- The load profile is obtained by scaling the Commercial Reference Building's annual energy consumption by the ratio of the actual building's square footage (**49,978 sq ft**) and the reference building's square footage (**53,628 sq ft**).
- The modeled load is highest on average during summer months but shows spikes in peak demand in the winter. Weekend loads are minimal.
- On weekdays, load begins to ramp up between **6:00 and 7:00 a.m.**, starts to decline around **5:00 p.m.**, and reaches a minimal consumption level by **9:00 p.m.**

	Medium Office Profile
Peak load	304.3 kW
Average load	103.5 kW
Minimum load	18.9 kW
Annual consumption	906,343 kilowatt-hours



Critical Load Modeling

- Three critical load scenarios were modeled for the resilience analysis:
 - 50% critical load:** Assumes 50% of the typical electric load must be met at every time step during a grid outage. All technologies are optimally sized for both outage survival and grid-connected operation.
 - 20% critical load:** Assumes 20% of the typical electric load must be met at every time step during a grid outage. All technologies are optimally sized for both outage survival and grid-connected operation.
 - Detailed model:** Defines a critical load profile using a bottom-up estimate of essential equipment power consumption. Operation is limited to July and August between 12 p.m. and 6 p.m. to provide a lower upfront capital cost resilience solution that does not rely on an emergency backup generator. The minimum required PV and battery sizes to survive the outage are reported.

Critical Load Type	Power (watts [W])	Quantity (store)	Quantity (school)	Start Hour	End Hour	Start Month	End Month
Lighting: LED A19 100 W Equivalent	14	5	5	12:00 p.m.	6:00 p.m.	July	August
Plug Load: Phone Charging	12	15	30	12:00 p.m.	6:00 p.m.	July	August
Plug Load: Refrigeration	500	3	0	12:00 p.m.	6:00 p.m.	July	August
HVAC: Airconditioning	1,000	4	4	12:00 p.m.	6:00 p.m.	July	August
HVAC: Fans	100	4	4	12:00 p.m.	6:00 p.m.	July	August

Utility Rate Summary

- Greenville takes service under the Tri-County Electric Cooperative.
- This analysis assumes **General Single-Phase Service** though **Rate Schedule 20**, based on utility bills provided for the Greenville Elementary School.
- The Tri-County Electric Cooperative offers **net energy metering (NEM)** for energy exported up to the site's annual consumption. Excess generation beyond site load is compensated at Seminole Electric Cooperative's **avoided costs**.

Schedule 20	Value
Basic facility charge	\$23/month
Energy charge	\$0.135/kilowatt-hour
Minimum monthly charge	\$30

Key Assumptions

Additional technology assumptions are listed in the appendix.

Inputs	Assumptions
Technologies evaluated	PV, wind turbines, battery energy storage, diesel generators
Ownership model	County-owned
Analysis period	25 years
Discount rate (nominal)	6.24%
Cost escalation rates (nominal)	1.66%/year (electricity); 1.97% (diesel fuel)
Tax rate	0%
Space available	Grocery store: 5,430 sq ft (50% of the total square footage); elementary school: 33 acres
PV costs	Rooftop PV (grocery store): \$2,232/kilowatts direct current (kWdc; 12- to 100-kW systems); \$1,920/kWdc (101- to 2,000-kW systems) Ground-mount PV (school): \$2,902/kWdc (12- to 100-kW systems); \$2,208/kWdc (101- to 2,000-kW systems)
Wind costs	\$7,692/kW (0- to 20-kW systems); \$5,776/kW (21- to 100-kW systems)
Battery costs	\$489/kW + \$452/kWh + \$5,926 fixed install costs
Diesel generator costs	\$650/kW (\$2.25/gallon fuel cost)
Incentives	0% and 30% federal investment tax credit (ITC) sensitivity for PV and wind and 30% ITC for battery storage: To be eligible, PV and wind projects must begin construction by July 4, 2026 (deadline passed) or projects must be placed in service by December 31, 2027 , regardless of start time. Storage projects must begin construction by December 31, 2033 , to qualify for 100% of the credit.

Sources: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>
https://www.cebn.org/media_resources/house-republicans-advanced-their-tax-reform-package-through-the-ways-and-means-committee/

Results

Scenarios Evaluated

- The following scenarios were evaluated separately for the Greenville Grocery Store and the Greenville Elementary School based on priorities discussed with the site:
 - **Business-as-usual:** The site purchases all electricity from the grid.
 - **Cost-optimal:** PV, wind turbines, and battery energy storage are optimally sized to minimize the site's total life cycle costs over a **25-year analysis period**.
 - **On-site energy generation:** PV, wind turbines, and battery energy storage are optimally sized to meet **20%, 40%, 60%, 80%, and 100%** of the site's annual electric load with on-site generation, subject to space availability. Rather than generating energy on-site to match load at every time step, the site may buy electricity from the grid during periods of low generation and offset these purchases by exporting excess generation during other periods. As a result, grid purchases may still occur even in the 100% scenario.
 - **Resilience:** PV, wind turbines, battery energy storage, and backup diesel generators are sized to meet **4-hour, 12-hour, 24-hour, 72-hour, and 168-hour** outages in **three critical load scenarios**.
 - If a diesel generator is selected, the minimum size is set to the maximum critical load.
 - When a PV system is sized independently to pair with a diesel generator, a **30-minute battery with an inverter capacity at the maximum critical load** is added to support tandem PV and generator operation and prevent excessive wear on the generator.

Cost-Optimal Results: Greenville Grocery Store

	BAU	Cost-Optimal: No PV/Wind ITC	Cost-Optimal: With PV/Wind ITC
PV size (kWdc)	–	54.3	54.3
Wind system size (kW)	–	0	0
Battery size (kW/kWh)	– / –	0 / 0	0 / 0
Year 1 utility cost	\$72,775	\$62,609	\$62,609
Year 1 utility bill reduction (%)	–	14%	14%
Year 1 operating costs	\$0	\$1,412	\$1,412
Upfront capital costs after incentives	\$0	\$121,198	\$86,952
Total life cycle costs	\$1,078,549	\$1,071,974	\$1,037,728
Net present value	–	\$6,575	\$40,821
On-site generation percent (%)	0%	14%	14%
Estimated life cycle emissions (tonnes of CO ₂ e)	3,740	3,391	3,391
Life cycle emissions reduction (%)	–	9%	9%

Only PV is cost-optimal for reducing the site's electricity costs over the analysis period under the listed assumptions.

Rooftop PV is maximized based on space available and can meet 14% of the site's annual electricity demand with renewable energy, regardless of ITC availability.

The site can reduce upfront capital costs by approximately 30% if PV systems are procured before the ITC expires, corresponding to a \$34,246 increase in net present value.

Cost-Optimal Results: Greenville Elementary School

	BAU	Cost Optimal: No PV/Wind ITC	Cost Optimal: With PV/Wind ITC
PV size (kWdc)	–	651.6	651.6
Wind system size (kW)	–	0	0
Battery size (kW/kWh)	– / –	0 / 0	0 / 0
Year 1 utility cost	\$122,632	\$333	\$333
Year 1 utility bill reduction (%)	–	99.7%	99.7%
Year 1 operating costs	\$0	\$13,032	\$13,032
Upfront capital costs after incentives	\$0	\$1,438,778	\$1,032,764
Total life cycle costs	\$1,817,445	\$1,655,082	\$1,249,068
Net present value	–	\$162,363	\$568,377
On-site generation percent (%)	0%	100%	100%
Estimated life cycle emissions (tonnes of CO ₂ e)	6,042	1,851	1,851
Life cycle emissions reduction (%)	–	69%	69%

Similarly, only PV is cost-effective at the elementary school. Ground-mount PV is sized to the NEM limit, serving 100% of the estimated community center load with renewable energy.

The site can reduce upfront capital costs by approximately 30% if PV systems are procured before the ITC expires, corresponding to a \$406,014 increase in net present value.

On-Site Generation

Results: No PV/Wind ITC

Because of space constraints at the grocery store, additional on-site generation scenarios are presented only for the [Greenville Elementary School](#).

Although a PV system capable of generating 100% of the site's load on an annual energy basis is cost-optimal, the site may choose to install capacity in smaller increments based on available upfront capital.

Greenville Elementary School	BAU	Cost-Optimal	20% On-Site Generation	40% On-Site Generation	60% On-Site Generation	80% On-Site Generation	100% On-Site Generation
PV size (kWdc)	–	651.6	130.3	260.6	391.0	521.3	651.6
Wind system size (kW)	–	0	0	0	0	0	0
Battery size (kW/kWh)	– / –	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
Year 1 utility cost	\$122,632	\$333	\$98,162	\$73,697	\$49,237	\$24,784	\$333
Year 1 utility bill reduction (%)	–	99.7%	20.0%	39.9%	59.8%	79.8%	99.7%
Year 1 operating costs (%)	\$0	\$13,032	\$2,606	\$5,213	\$7,819	\$10,426	\$13,032
Upfront capital costs after incentives	\$0	\$1,438,778	\$287,756	\$575,511	\$863,267	\$1,151,023	\$1,438,778
Total life cycle costs	\$1,817,445	\$1,655,082	\$1,784,827	\$1,752,260	\$1,719,794	\$1,687,418	\$1,655,082
Net present value	–	\$162,363	\$32,618	\$65,185	\$97,651	\$130,028	\$162,363
On-site generation percent (%)	0%	100%	20%	40%	60%	80%	100%
Estimated life cycle emissions (tCO ₂ e)	6,042	1,851	5,204	4,366	3,528	2,689	1,851
Life cycle emissions reduction (%)	–	69%	14%	28%	42%	55%	69%

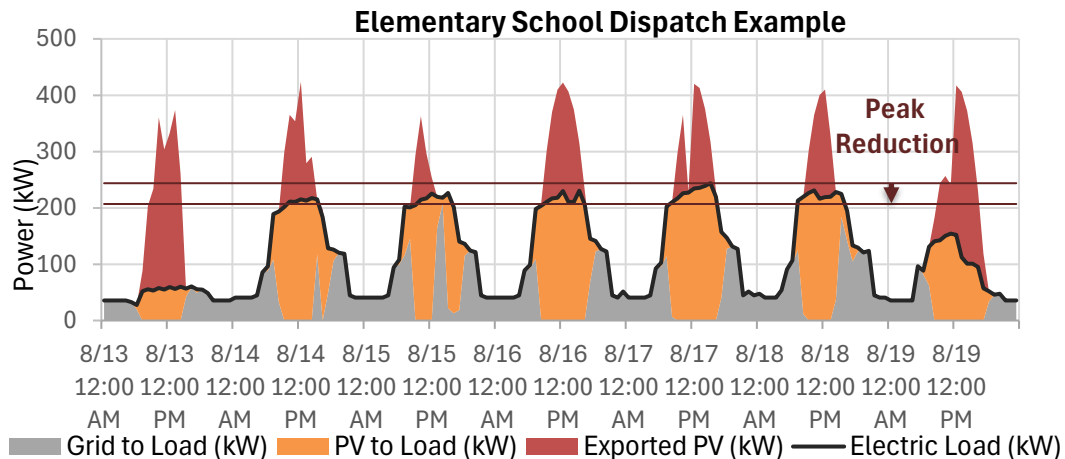
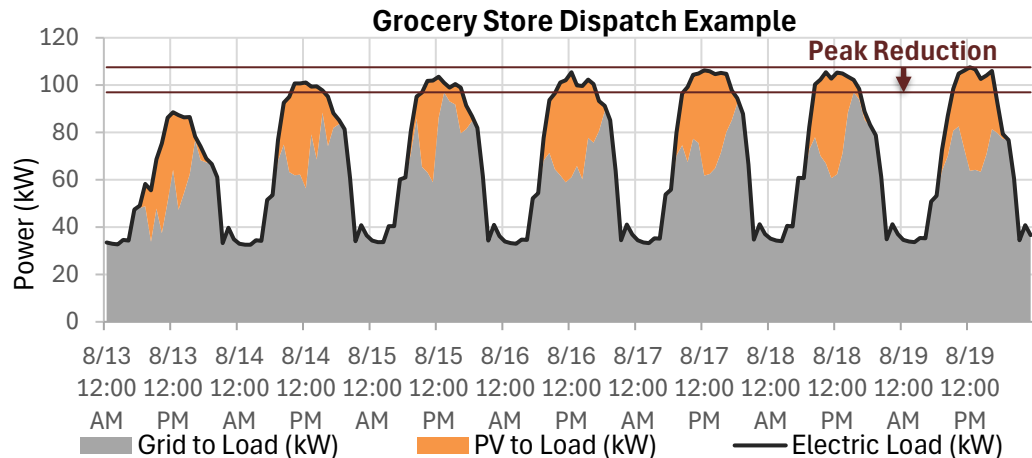
On-Site Generation Results: With PV/Wind ITC

If PV systems are procured at the **Greenville Elementary School** before the ITC expires, upfront capital costs decrease by approximately **30%** across all scenarios, corresponding to a **3.5x** increase in net present value.

Greenville Elementary School	BAU	Cost-Optimal	20% On-Site Generation	40% On-Site Generation	60% On-Site Generation	80% On-Site Generation	100% On-Site Generation
PV size (kWdc)	–	651.6	130.3	260.6	391.0	521.3	651.6
Wind system size (kW)	–	0	0	0	0	0	0
Battery size (kW/kWh)	– / –	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
Year 1 utility cost	\$122,632	\$333	\$98,162	\$73,697	\$49,237	\$24,784	\$333
Year 1 utility bill reduction (%)	–	99.7%	20.0%	39.9%	59.8%	79.8%	99.7%
Year 1 operating costs (%)	\$0	\$13,032	\$2,606	\$5,213	\$7,819	\$10,426	\$13,032
Upfront capital costs after incentives	\$0	\$1,032,764	\$206,553	\$413,106	\$619,658	\$826,211	\$1,032,764
Total life cycle costs	\$1,817,445	\$1,249,068	\$1,703,624	\$1,589,854	\$1,476,186	\$1,362,606	\$1,249,068
Net present value	–	\$568,377	\$113,821	\$227,591	\$341,260	\$454,839	\$568,377
On-site generation percent (%)	0%	100%	20%	40%	60%	80%	100%
Estimated life cycle emissions (tCO ₂ e)	6,042	1,851	5,204	4,366	3,528	2,689	1,851
Life cycle emissions reduction (%)	–	69%	14%	28%	42%	55%	69%

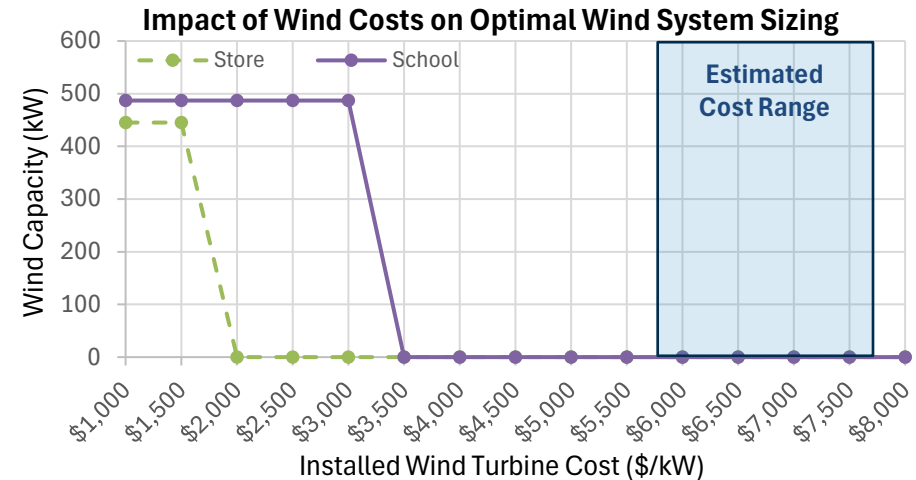
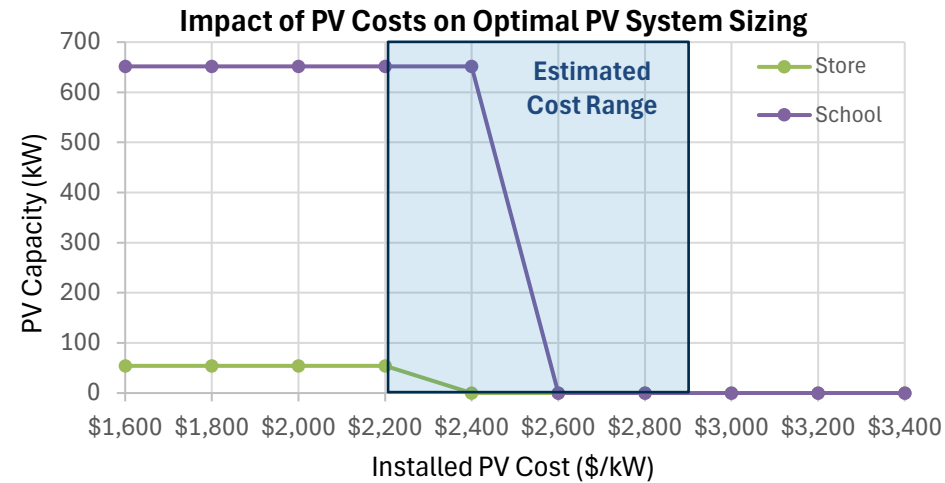
Dispatch Example

- Scenario: Cost-optimal.
- PV reduce site load, but the optimized system size is large enough only to generate excess energy to export back to the grid at the Greenville Elementary School.
- The average peak load reduction is **6.8 kW/month** at the grocery store and **26.3 kW/month** at the elementary school.



Impact of Technology Cost on Optimal Sizing

- To address uncertainties with cost for new PV and wind energy projects, optimal system sizes are shown across a range of technology installed costs (no ITC).
 - For PV cost sensitivity scenarios, only PV is sized (wind and battery storage systems are not considered). Similarly, only wind turbines are sized in the wind cost sensitivity scenarios.
 - All scenarios assume a **NEM limit** equal to the site's annual electricity consumption.
- At the Greenville Elementary School, optimal technology sizes level off at the NEM limit at the lower end of the cost range.
- At the Greenville Grocery Store, optimal PV capacity is constrained by available space. Wind results for the grocery store are shown with dashed lines because the model allows capacity to be sized up to the NEM limit, but wind installations are not feasible because of space constraints.



Resilience Results: Greenville Grocery Store

PV/wind ITC increases net present value but does not affect optimal system sizing.

When not constrained to minimize upfront capital costs, PV is sized to the space limit, independent of critical load, to maximize grid-connected savings.

Total life cycle costs do not include the cost of fuel, which depends on both outage timing and duration.

Resilience increases life cycle costs for the site but could **reduce outage-related expenses** and serve as a community resource. These benefits are not explicitly modeled here.

	No PV/Wind ITC			With PV/Wind ITC		
Critical load profile	50%	20%	Detailed	50%	20%	Detailed
PV size (kWdc)	54.3	54.3	12.5	54.3	54.3	12.5
Wind system size (kW)	0	0	0	0	0	0
Battery size (kW/kWh)	58 / 29	23 / 12	13 / 51	58 / 29	23 / 12	13 / 51
Generator size (kW)	58	23	0	58	23	0
Year 1 utility cost	\$62,608	\$62,608	\$70,437	\$62,608	\$62,608	\$70,437
Year 1 utility bill reduction (%)	14%	14%	3.2%	14%	14%	3.2%
Year 1 operating costs	\$3,768	\$2,443	\$1,211	\$3,768	\$2,443	\$1,211
Upfront capital costs after incentives	\$193,269	\$152,578	\$53,290	\$159,023	\$118,332	\$45,419
Total life cycle costs	\$1,182,254	\$1,120,079	\$1,116,819	\$1,148,008	\$1,085,833	\$1,108,948
Net present value	-\$103,705	-\$41,530	-\$38,270	-\$69,459	-\$7,284	-\$30,399
On-site generation percent (%)	14.0%	14.0%	3.2%	14.0%	14.0%	3.2%
Estimated life-cycle emissions (tCO ₂ e)	3,390	3,390	3,659	3,390	3,390	3,659
Life-cycle emissions reduction (%)	9%	9%	2%	9%	9%	2%

Resilience Results: Greenville Elementary School

PV/wind ITC increases net present value
but does not affect optimal system sizing.

	No PV/Wind ITC			With PV/Wind ITC		
Critical load profile	50%	20%	Detailed	50%	20%	Detailed
PV size (kWdc)	651.6	651.6	9.9	651.6	651.6	9.9
Wind system size (kW)	0	0	0	0	0	0
Battery size (kW/kWh)	152 / 76	61 / 30	11 / 45	152 / 76	61 / 30	11 / 45
Generator size (kW)	152	61	0	152	61	0
Year 1 utility cost	\$333	\$333	\$120,764	\$333	\$333	\$120,764
Year 1 utility bill reduction (%)	99.7%	99.7%	1.5%	99.7%	99.7%	1.5%
Year 1 operating costs	\$18,943	\$15,485	\$992	\$18,943	\$15,485	\$992
Upfront capital costs after incentives	\$1,619,958	\$1,513,802	\$44,714	\$1,213,951	\$1,107,790	\$38,520
Total life cycle costs	\$1,932,130	\$1,769,895	\$1,850,561	\$1,526,124	\$1,363,884	\$1,844,367
Net present value	-\$114,685	\$47,551	-\$33,115	\$291,322	\$453,562	-\$26,921
On-site generation percent (%)	100%	100%	1.5%	100%	100%	1.5%
Estimated life cycle emissions (tCO ₂ e)	1,851	1,851	5,978	1,851	1,851	5,978
Life cycle emissions reduction (%)	69%	69%	1%	69%	69%	1%

When not constrained to minimize upfront capital costs, PV is sized to the NEM limit, independent of critical load, to maximize grid-connected savings.

If the optimally sized PV system is procured, resilience can be achieved at a positive net present value at 20% critical load without the ITC and at both 50% and 20% critical load with ITC.

Conclusion

Conclusion and Next Steps

- Based on the analysis assumptions, only PV is cost-effective to reduce electricity costs at the Greenville Grocery Store and Elementary School over a 25-year analysis period.
 - A **54.3-kWdc** rooftop PV system is cost-optimal at the grocery store, serving **14%** of annual load with on-site generation. A **651.6-kWdc** ground-mount PV system is cost-optimal at the elementary school, serving **100%** of the annual load.
- Meeting the modeled 4-hour to 1-week outages with PV, wind, and battery storage alone is not cost-effective. Instead, in the 50% and 20% critical load scenarios, the presented solution includes a diesel generator sized to the maximum critical load and adds a 30-minute battery with matching inverter capacity to support tandem PV-generator operation.
 - This configuration results in negative NPV at the grocery store across all scenarios. At the elementary school, NPV is positive when ITC is available for PV and remains positive at 20% critical load even without the ITC.
 - To avoid incurring battery costs, the site may consider relying solely on diesel generation during grid outages. However, **generator reliability decreases as outage duration increases**, and generators require **regular testing and maintenance and a reliable fuel supply** during emergencies. An estimate of generator O&M costs are included in this analysis, but fuel costs are not as they depend on the frequency and duration of actual outages.
 - This analysis does not include any avoided outage expenses and community resilience benefits, which could offset the modeled costs.
- An additional resilience scenario was evaluated to size technologies to support a limited critical load (primarily phone charging and space cooling) without relying on emergency backup generators and fuel resupply during extended outages. Although this approach significantly reduced upfront capital costs, it remained NPV-negative over the analysis period.

Thank You

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Appendix

Detailed Assumptions

Financial Assumptions

Inputs	Assumption
Technologies evaluated	PV, wind turbines, battery energy storage, diesel generators
Model objective	Minimize the total life cycle cost of electricity (includes the present value of grid electricity purchases and the capital, installation, replacement, and O&M costs of any new distributed energy technologies)
Ownership model	County-owned
Analysis period	25 years
Operations and maintenance (O&M) cost escalation rate	2.5%/year
Discount rate (nominal)	6.24%
Electricity cost escalation rate (nominal)	1.66%/year
Generator fuel cost escalation rate (nominal)	1.97%/year
Tax rate	0%

Source: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>

PV Assumptions

Inputs	Assumptions
System type	Grocery store: rooftop, standard module Elementary school: ground-mount, fixed tilt, standard module
Technology resource	Typical meteorological year weather data from the National Solar Resource Database (capacity factor = 16.7%)
Inverter efficiency	96%
System losses	14%
Tilt	20°
Azimuth	180°
DC to AC size ratio	1.2
Space available	Unlimited
Total installed costs	Rooftop: \$2,232/kWdc (12- to 100-kW systems); \$1,920/kWdc (101- to 2,000-kW systems) Ground-mount: \$2,902/kWdc (12- to 100-kW systems); \$2,208/kWdc (101- to 2,000-kW systems)
O&M costs	\$26/kW/year (12- to 100-kW systems); \$20/kW/year (101- to 2,000-kW systems)
Incentives	None

Source: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>

Wind Assumptions

Inputs	Assumptions
System type	Small-scale (0–20 kW) and commercial-scale (21–100 kW) turbines
Technology resource	Typical meteorological year weather data from the NLR Wind Integration National Dataset (WIND) Toolkit (capacity factor = 13.8%–21.2%)
Hub height	20 meters (small scale); 40 meters (commercial scale)
Space available	Unlimited
Total installed costs	\$7,692/kW (small-scale); \$5,776/kW (commercial-scale)
O&M costs	\$42/kW/year
Incentives	None

Source: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>

Battery Storage Assumptions

Inputs	Assumptions
System type	Lithium-ion battery
Rectifier and inverter efficiencies	96%
DC-to-DC roundtrip efficiency	97.5%
AC-to-AC roundtrip efficiency	89.9%
Minimum state of charge	20%
Can the grid charge the battery?	Yes
Total installed costs	\$452/kWh + \$489/kW + \$5,926
O&M costs	2.5% of installed costs per year; includes system replacement costs
Incentives	30% ITC

Source: <https://reopt.nrl.gov/tool/reopt-user-manual.pdf>

Generator Assumptions

Inputs	Assumptions
System type	Diesel generator
Electrical efficiency	32.2%
Fuel cost (\$/gallon)	\$2.25
Fuel available (gallons)	Unlimited
Total installed costs	\$650/kW
O&M costs	\$20/kW/year
Incentives	None

Source: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>

Resilience Modeling

Inputs	Assumptions
Critical load scenarios	50% of typical load, 20% of typical load, bottom-up modeling based on devices.
Modeled outage scenarios	Technologies are optimally sized to meet 4-hour, 12-hour, 24-hour, and 72-hour outages with at least a 90% probability of survival, regardless of when the outage occurs during the year. When a PV system is sized independently to pair with a diesel generator, a 30-minute battery with an inverter capacity at the maximum critical load is added to support tandem PV and generator operation and prevent excessive wear on the generator. A scenario for a 1 outage (168 hours) is also evaluated. However, as outage duration increases, generator reliability declines, reducing the probability of survival at 168 hours to approximately 85%.

Source: <https://reopt.nlr.gov/tool/reopt-user-manual.pdf>